A Prospective, Randomized, Double-Blind, Crossover Comparison Evaluating Compatibility and Safety of LY900014 and Insulin Lispro with an External Continuous Subcutaneous Insulin Infusion System in Adult Patients with Type 1 Diabetes (PRONTO-Pump)

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LY900014

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Protocol Electronically Signed and Approved by Lilly: 23 October 2017
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1. Synopsis

Title of Study:
A Prospective, Randomized, Double-Blind, Crossover Comparison Evaluating Compatibility and Safety of LY900014 and Insulin Lispro with an External Continuous Subcutaneous Insulin Infusion System in Adult Patients with Type 1 Diabetes (PRONTO-Pump)

Rationale:
Insulin pumps use only rapid-acting insulin for both basal and bolus insulin requirements. Basal insulin is a continuous infusion of insulin that is delivered automatically 24 hours a day. The purpose of basal insulin is to cover hepatic glucose production and maintain glucose stability during fasting states (between meals and during sleep). Bolus insulin is delivered "on-demand," by the patient, for food intake and/or to correct glucose levels that are above the patient's target range, delivered separately or together.

Because insulin pump therapy uses only rapid-acting insulin, the onset of diabetic ketoacidosis (DKA) can occur quickly if insulin delivery is interrupted for a period of time. The most important DKA prevention strategies are: 1) adhering to a routine blood glucose (BG) monitoring/continuous glucose monitoring schedule and 2) never ignoring an unexplained high BG. A high BG level that is not responding to a correction bolus via insulin pump may indicate an infusion set occlusion or insulin pump malfunction.

The aim of this study is to compare LY900014 and insulin lispro with respect to the rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with BG >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump.

Objective(s)/Endpoints:

Objectives	Endpoints
Primary Objective	
1. To compare LY900014 and insulin lispro with respect to the rate of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump	1. Rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump, during the 6-week treatment period

Objectives	Endpoints
Secondary Objectives	
2. To compare LY900014 and insulin lispro with respect to the incidence of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump	2. Incidence (percent of patients with at least 1 event) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump, during the 6-week treatment period
3. To compare LY900014 and insulin lispro with respect to the rate and incidence of premature infusion set changes	3. Rate (events/patient/30 days) and incidence of premature infusion set changes by reason (infusion set kinked, came out, or leaking; empty pump reservoir; infusion site pain or redness; pump occlusion alarm; suspected infusion set occlusion; other) during the 6-week treatment period
4. To compare LY900014 and insulin lispro with respect to the time interval until infusion set change	4. Time interval until infusion set change during the 6-week treatment period
5. To compare LY900014 and insulin lispro with respect to total, basal, and bolus insulin dose	5. Bolus/total insulin dose ratio at the end of the 6-week treatment period
6. To compare LY900014 and insulin lispro with respect to the interstitial glucose reduction rate from hyperglycemia following a non-meal-related correction bolus delivered via the pump	6. Interstitial glucose reduction rate (glucose reduction [mg/dL and mmol/L] per minute) within 4 hours following a non-meal-related correction bolus via the pump, from hyperglycemia (interstitial glucose >180 mg/dL [10 mmol/L]) to recovery (interstitial glucose ≤180 mg/dL), from up to 6 weeks of CGM use
7. To compare LY900014 and insulin lispro with respect to the rate of severe hypoglycemic events	7. Rate (events/patient/100 years) of severe hypoglycemic events during the 6-week treatment period

Abbreviations: CGM = continuous glucose monitoring; SMBG = self-monitored blood glucose.

Summary of Study Design:

Study I8B-MC-ITSI is a Phase 3, prospective, randomized, double-blind, outpatient, multinational, multicenter, 2-treatment group, crossover, active-controlled study conducted in patients with type 1 diabetes currently using an external continuous subcutaneous insulin infusion pump. In the two treatment groups, LY900014 and insulin lispro, meal bolus doses will be delivered immediately prior to each meal (0 to 2 minutes) in a double-blind manner.

Treatment Arms and Duration:

The study includes a 1-week screening period and a 2-week lead-in period followed by a 2-period crossover and a 4-week post-treatment safety follow-up. Each period of the crossover will consist of 6 weeks of treatment with no washout between periods.

Patients will be randomized to 1 of the 2 treatment sequences in a 1:1 ratio:

Sequence A: LY900014 → insulin lispro
Sequence B: insulin lispro → LY900014

Number of Participants:

Approximately 60 participants will be screened to achieve 48 randomized patients and approximately 42 patients completing 12 weeks of treatment.

Statistical Analysis:

Safety analyses will be conducted on the Safety Population. Analyses of AEs (including DKA) will include all data collected during the course of the entire 6-week treatment period for each treatment group regardless of IP use. Analyses of hypoglycemia will be conducted on data collected prior to permanent discontinuation (ie, last dose) of IP in each 6-week treatment period. Unless otherwise specified, pump-related safety analyses (infusion set failures, premature infusion set changes, time interval until infusion set change, and interstitial glucose reduction rate) will exclude data (if any) that are collected while patients temporarily are off pump or off IP. Data collected during the safety follow-up period will not be used for comparisons between treatment groups.

Unless otherwise noted, all tests of treatment effects will be conducted at a 2-sided alpha level of 0.05 and confidence intervals (CIs) will be calculated at 95%, 2-sided.

Baseline is defined as the last non-missing measurement at or before the randomization visit (Visit 3), unless otherwise specified.

The primary objective of this study is to compare LY900014 and insulin lispro with respect to the rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump, during the 6-week treatment period.

To be considered as the primary endpoint, an infusion set failure must result in a premature infusion set change due to:

1. pump occlusion alarm(s), OR
2. unexplained hyperglycemia that meets the following conditions:
   - Associated with a non-meal-related correction bolus delivered via the pump at least 1 hour before the infusion set change
   - Most recent self-monitored blood glucose (SMBG) within 1 hour before the correction bolus >250 mg/dL (13.9 mmol/L)
   - At least 1 SMBG obtained following the correction bolus prior to the infusion set change
   - Within 1 hour (+10 minutes) following the correction bolus, the most recent SMBG prior to the infusion set change does not indicate a decrease in blood glucose

If more than 1 non-meal-related correction bolus is given for an unexplained hyperglycemia, only the first correction bolus time will be used to determine the 1-hour (+10 minutes) timeframe following the correction bolus.

The analyses of the primary objective will be conducted on the Safety Population including data collected prior to permanent discontinuation of investigational product (ie, last dose) in each randomized treatment period (Period I and Period II) and excluding data (if any) that are collected while patients temporarily are off pump or off IP. Treatment group comparisons will be performed using Wilcoxon signed-rank test at the full significance level of 0.05, with patients who are dosed in both treatment periods. No multiplicity adjustments will be made for the analysis of secondary and exploratory objectives.

The incidence (percent of patients with at least 1 event) of infusion set failure as defined in the primary objective, premature infusion set changes by reason, and severe hypoglycemia will be analyzed using Prescott's exact test.

The Wilcoxon signed-rank test will be used to analyze the rate of premature infusion set changes by reason, and severe hypoglycemia, as described above for the primary endpoint.

For continuous measures, summary statistics will include sample size, mean, standard deviation, median, minimum, and maximum for both the actual and the change from baseline measurements. Least square (LS) means and standard errors derived from the analysis models will also be displayed for the change from baseline measurements. Treatment comparisons will be displayed showing the treatment difference LS means and the 95% CIs for the treatment differences, along with the p-values for the treatment comparisons.

For categorical measures (such as incidence of adverse events), summary statistics will include sample size, frequency, and percentages. Prescott's test will be used for treatment comparisons, unless otherwise specified.

2. Schedule of Activities

Table ITSI.1. Schedule of Activities

Study Procedure	Study Screening	Lead-in Period	Study Period I				Study Period II			Safety Follow-up	ED
eCRF Visit Number	**1**	**2**	**3**	**4**	**5[a]**	**6**	**7**	**8[a]**	**9**	**801**	**ED**
Visit Window (± days)	**0**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**7**	
Time on Study Relative to First Active Treatment Dose (weeks)	**-3**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**16**	
Informed consent signed	X										
IWRS	X	X	X	X	X	X	X	X	X	X	X
Patient eligibility review	X										
Randomization[b]			X								
Clinical Assessments											
Patient demographics	X										
Medical history and preexisting conditions	X										
Physical exam/height	X										
Record pump model and reservoir size[c]	X										
Set pump bolus delivery speed to Standard (1.5 U/minute)		X									
Concomitant medication	X	X	X	X	X	X	X	X	X	X	X
Adverse events and product complaints[d]	X	X	X	X	X	X	X	X	X	X	X
Train patient on use of eCOA to enter all infusion set and reservoir changes and all non-meal-related correction boluses[e]		X									
Review/remind subjects of entering non-meal-related correction boluses			X	X	X	X	X	X	X		X
Review infusion set and reservoir changes, pump occlusion alarms, and suspected occlusions from eCOA[f]		X	X	X	X	X	X	X	X		X
Remind patient to monitor SMBG following all non-meal-related correction boluses for unexplained hyperglycemia[e]		X	X	X	X	X	X	X			
Body weight[g]	X	X	X	X		X	X		X	X	X
Vital signs: blood pressure/pulse rate[h]	X	X	X	X		X	X		X	X	X

Study Procedure	Study Screening	Lead-in Period	Study Period I				Study Period II			Safety Follow-up	ED
eCRF Visit Number	**1**	**2**	**3**	**4**	**5[a]**	**6**	**7**	**8[a]**	**9**	**801**	**ED**
Visit Window (± days)	**0**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**7**	
Time on Study Relative to First Active Treatment Dose (weeks)	**-3**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**16**	
ECG (12-lead)[i]	X										
Educate patient on adjusting pump basal rates and bolus calculator settings[j]		X	X	X		X	X		X		
Transfer to insulin lispro[k]		X									
Crossover to alternate IP and reset pump to basal rates and bolus calculator settings used at randomization						X					
Start non-study rapid-acting analog and reset pump to basal rates and bolus calculator settings used at randomization									X		X
Change infusion set and reservoir at investigative site[l]		X	X			X			X[m]		X[m]
Optimize basal rates and bolus calculator settings[n]		X	X	X		X	X		X		
Ancillary Supplies/Diaries/IP											
Dispense pump infusion sets and reservoirs		X	X	X		X	X		X		
Dispense eCOA, glucometer, ancillary supplies, pair glucometer with eDiary and complete training		X									
Dispense ancillary supplies as needed		X	X	X		X	X		X		
Review unexplained hyperglycemia troubleshooting		X									
Dispense blood ketone meter and strips and train patient on use.		X									
Dispense IP[o]		X	X	X		X	X				
Train on collecting 4-point SMBG profiles[p]		X									
Review 4-point SMBG profiles[q]			X	X	X	X	X	X	X	X	X

Study Procedure	Study Screening	Lead-in Period	Study Period I				Study Period II			Safety Follow-up	ED
eCRF Visit Number	**1**	**2**	**3**	**4**	**5[a]**	**6**	**7**	**8[a]**	**9**	**801**	**ED**
Visit Window (± days)	**0**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**7**	
Time on Study Relative to First Active Treatment Dose (weeks)	**-3**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**16**	
Review pump data for use in clinical decision making and data entry of required fields into Inform[r]		X	X	X		X	X		X		
Record in Inform total basal and total bolus units for last 3 days prior to visit		X	X	X		X	X		X	X	X
Record in Inform total number of meal-related bolus doses given as Normal, Square Wave, and Dual Wave during the 7 days prior to visit		X	X	X		X	X		X	X	X
Record in Inform pump basal rates, CRs, ISFs, and AIT from last day prior to visit		X	X			X			X		X
Record in Inform the breakfast CR and ISF, and the AIT from last day prior to visit		X	X	X		X	X		X		X
Record in Inform any blood ketone meter test results			X	X		X	X		X		X
Review/discuss blood ketone meter test results, if applicable			X	X		X	X		X		X
Review/discuss hypoglycemia data			X	X	X	X	X	X	X	X	X
Train patient on CGM including use of event markers to record all breakfast meal times (carb event) and corresponding meal bolus times (insulin event)[p]		X									
Review CGM use of event markers and corresponding meal times			X	X	X	X	X	X	X		X
Pair CGM transmitter and receiver		X									
Dispense CGM sensors		X	X	X		X	X				
Insert new CGM sensor and attach transmitter at site during visit.		X	X			X					
Synchronize the times of pump, eCOA,		X	X	X		X	X				

Study Procedure	Study Screening	Lead-in Period	Study Period I				Study Period II			Safety Follow-up	ED
eCRF Visit Number	**1**	**2**	**3**	**4**	**5[a]**	**6**	**7**	**8[a]**	**9**	**801**	**ED**
Visit Window (± days)	**0**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**7**	
Time on Study Relative to First Active Treatment Dose (weeks)	**-3**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**16**	
glucometer, ketone meter, and CGM receiver at site during visit											
Download and review CGM data for clinical decision making and transmit to secure cloud storage			X	X		X	X		X		X
Remind patient to change CGM sensor every 7 days		X	X	X	X	X	X	X			
Remove last CGM sensor, detach transmitter, and return transmitter and receiver to site									X		X
Drug accountability			X	X		X	X		X		X
Return eCOA devices										X	X
Patient returns used and unused study drug supplies (used and unused vials)			X	X		X	X		X		X
Laboratory Assessments											
Urinalysis panel	X										
Pregnancy test[s]	X		X								
Follicle-stimulating hormone test[t]	X										
Chemistry panel	X		X								
Hematology	X		X								
1.5-Anhydroglucitol			X			X			X		X
Hemoglobin A1c	X		X			X			X	X	X
Anti-insulin lispro antibodies		X	X			X			X	X	X

Abbreviations: AIT = active insulin time; CGM = continuous glucose monitoring; BG = blood glucose; CR = carb ratio; ECG = electrocardiogram; eCOA = electronic clinical outcomes assessment, eCRF = electronic case report form; ED = early discontinuation; IP = investigational product; ISF = insulin sensitivity factor; IWRS = interactive web response system; SMBG = self-monitored blood glucose.

[a] Telephone visits are indicated by shaded columns. Activities include: record visit in the IWRS; collect adverse events and concomitant medications; review of pump occlusion alarms and suspected occlusions; review BG readings and study drug doses; adjust CRs, ISFs, AIT and basal rates as needed; review hypoglycemic events; provide reminders regarding scheduled visits and SMBG profiles, as applicable.

[b] Randomization should occur after completion of all Visit 3 procedures.
[c] The MiniMed Mio infusion set with 6-mm length cannula and 23" tubing length will be provided for all patients to use during the study.
[d] Any insulin pump issue will be assessed at the investigative site and if not resolved the investigator should report the complaint directly to the pump manufacturer in accordance with the product labeling.
[e] Non-meal-related correction boluses are not associated with a meal or combined with a meal bolus.
[f] Infusion set and reservoir changes must be at same time.
[g] Patients should be advised to remove their shoes/coats and empty their pockets before the body weight is obtained.
[h] Vital sign measurements should be determined after patients have been seated quietly for at least 5 minutes in a chair with feet on the floor. The arm used for blood pressure measurement should be supported at heart level.
[i] Patients must be supine for approximately 5 to 10 minutes before ECG collection and remain supine but awake during ECG collection.
[j] Initial training at Visit 2 will include diabetes education, insulin pump education, and nutrition counseling. Training will include a review of correct pump use (frequency of reservoir and infusion set changes, infusion site rotation, carb counting, and adjusting pump basal rates and bolus calculator settings) and review patient's ability to trouble shoot unexplained hyperglycemia and treat hypoglycemia. Appropriate site personnel will administer training and education using locally approved diabetes education/training materials and programs or by using other materials that may be provided by the sponsor. Patients may be provided abbreviated training and education at visits following Visit 2 based upon patient needs.
[k] Only for patients using insulin aspart or insulin glulisine.
[l] Complete all study procedures prior to reservoir and infusion set change at Visits 2, 3, 6, and 9. Under the observation of site staff, patients will fill a new pump reservoir and infusion set, insert the infusion set cannula into an appropriate pump site, and begin infusion.
[m] At Visit 9 or ED, start non-study rapid-acting insulin.
[n] Investigators, in discussion with patients, adjust basal rates and CRs, ISFs, and AIT for calculating bolus and correction doses to meet the target BG levels.
[o] Complete all study procedures prior to initiating IP at Visits 2, 3, and 6.
[p] Training may be repeated at other visits, as needed.
[q] Patients should be instructed to perform a minimum of 4-point SMBG profiles daily starting at Visit 2.
[r] Enter into Inform current pump basal rates, CRs, ISFs, and AIT from Device Settings Snapshot; total basal units and total bolus units for last 3 days prior to visit, and total number of Normal, Square Wave, and Dual Wave bolus doses given during the last 7 days prior to the visit from Daily Detail report.
[s] Serum pregnancy test must be performed in women of childbearing potential at Visit 1 followed by a urine pregnancy test within 24 hours prior to IP exposure at randomization (Visit 3) and at other times at the investigator's discretion. When required per local regulations and/or institutional guidelines, local pregnancy testing will occur at mandatory times during the study treatment period.
[t] Follicle-stimulating hormone test must be performed at Visit 1 for a postmenopausal woman who is between 50 and 54 years of age (inclusive) with an intact uterus, not on hormone therapy, and has had at least 6 months of spontaneous amenorrhea.

3. Introduction

3.1. Study Rationale

Prandial insulin with faster onset and/or faster offset characteristics might reduce glycemic excursions, time to recovery from hyperglycemia, and the incidence of delayed postprandial hypoglycemia compared to currently available rapid-acting insulin analogs. Rapid-acting insulins, such as Humalog®, have been shown to have a more rapid onset of action compared to human insulin; however, the general consensus is that they are not rapid enough to match carbohydrate absorption, whether delivered by pump or syringe/pen injector, limiting efficacy. An ultra-rapid-acting prandial insulin that would shift the pharmacokinetic and pharmacodynamic profiles of insulin to provide an even faster onset of action would better match carbohydrate absorption and allow for efficacious dosing immediately prior to meals. The time-action profile of a rapid-acting insulin could be enhanced through the addition of excipients to an existing formulation to increase capillary blood flow and/or enhance vascular permeability. An ultra-rapid insulin would be useful in the treatment of type 1 diabetes mellitus (T1DM) and type 2 diabetes mellitus (T2DM) when delivered by multiple daily injections (MDI), continuous subcutaneous insulin infusion (CSII), or in the development of closed loop insulin delivery systems.

Insulin pumps use only rapid-acting insulin for both basal and bolus insulin requirements. Basal insulin is a continuous infusion of insulin that is delivered automatically 24 hours a day. The purpose of basal insulin is to cover hepatic glucose production and maintain glucose stability during fasting states (between meals and during sleep). Bolus insulin is delivered "on-demand," by the patient, for food intake and/or to correct glucose levels that are above the patient's target range, delivered separately or together.

Because insulin pump therapy uses only rapid-acting insulin, the onset of diabetic ketoacidosis (DKA) can occur quickly if insulin delivery is interrupted for a period of time. DKA develops when insulin levels are insufficient to meet the body's basic metabolic requirements. DKA is an acute metabolic complication of diabetes characterized by hyperglycemia, hyperketonemia, and metabolic acidosis. Signs and symptoms may include vomiting, abdominal pain, deep gasping breathing, increased urination, weakness, confusion, and occasionally; loss of consciousness. The most important DKA prevention strategies are: 1) adhering to a routine blood glucose (BG) monitoring/continuous glucose monitoring (CGM) schedule and 2) never ignoring an unexplained high BG. A high BG that is not responding to a correction bolus via insulin pump may indicate an infusion set occlusion or insulin pump malfunction.

The aim of this study is to compare LY900014 and insulin lispro with respect to the rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump.

3.2. Background

There have been many advances in the treatment of T1DM in the last 20 years; however, reaching and maintaining glycemic goals remains challenging even under intensive insulin therapy regimens. Only approximately 30% of insulin-requiring patients with diabetes are able to reach the goal hemoglobin A1c (HbA1c) target of <7% (Stark Casagrande et al. 2013). Currently available rapid-acting insulin analogs continue to be unable to match the kinetics of physiological post-meal insulin secretion, which is biphasic. Thus, there remains a need to continue to develop formulations with a time-action profile that more closely approximates that of endogenous insulin section.

Ideally, currently available rapid-acting insulin analogs should be injected or bolused 10 to 15 minutes prior to meal consumption to control postprandial glucose (PPG). However, many people inject or bolus their rapid-acting insulin at the time of the meal or after the meal. According to survey data from the T1D Exchange on the timing of prandial insulin injection, 16.3% of patients indicated they inject 15 minutes prior to meals, 33% inject 1 to 14 minutes prior to meals, 38.6% inject at the time of the meal, and 12.1% inject after meals. Because patients often inject or bolus later than recommended, there is a greater mismatch between insulin action and postprandial BG elevations.

LY900014 is a formulation of insulin lispro developed as an ultra-rapid acting insulin with a faster onset of action and shorter duration of action compared to currently available rapid-acting insulin analogs. LY900014 contains the prostacyclin analog (treprostinil), citrate, and other excipients. This formulation involves the novel use of a microdose of treprostinil as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than an as an active pharmaceutical ingredient that elicits a systemic effect. Treprostinil is a prostacyclin analog for the treatment of symptomatic pulmonary arterial hypertension, and has been approved in the United States since 2002 (Remodulin package insert, 2014) and in Europe since 2005 (PMR [WWW]). Sodium citrate, an excipient that speeds insulin absorption, is also included in the formulation to further enhance the absorption of insulin lispro. Sodium citrate and the other excipients in the LY900014 formulation are listed in the Food and Drug Administration (FDA) Generally Recognized As Safe food additives database and in the FDA Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Nine LY900014 Phase 1 clinical studies have been conducted to date in healthy subjects or patients with T1DM or T2DM. Across these studies, LY900014 has consistently demonstrated a faster time-action profile than Humalog. In patients with T1DM or T2DM treated with MDI insulin therapy, LY900014 significantly reduced PPG excursions compared to Humalog when both were dosed by syringe at the start of a test meal. Two studies in patients with T1DM have been completed using CSII in an in-patient setting, both of which have demonstrated accelerated time action. LY900014 has been well tolerated in both healthy subjects and patients with diabetes mellitus. Notably, there have been no clinically significant increases in adverse events (AEs) associated with systemic absorption of treprostinil (headache, diarrhea, nausea, jaw pain,

vasodilatation, rash, edema, and hypotension are described in the Remodulin package insert, 2014). All samples of treprostinil from LY900014 administration were below the lower limit of quantitation. The Investigator's Brochure (IB) describes the clinical and nonclinical development of LY900014.

3.3. Benefit/Risk Assessment

In Phase 1 studies, the assessment of AEs, hypoglycemic events, local tolerability, vital signs, physical examination, electrocardiogram (ECG), anti-insulin lispro antibodies, and clinical laboratory assessments did not reveal any specific risks of LY900014 beyond those already known for insulin lispro.

More detailed information about the known and expected benefits and risks of insulin lispro may be found in the country-specific product labeling (for example, Patient Information Leaflet, Package Insert, or Summary of Product Characteristics).

Safety evaluation in this study will include hypoglycemia, treatment-emergent adverse event (TEAEs), serious adverse events (SAEs), clinical laboratory assessments, anti-insulin lispro antibodies, CGM, frequent visits, ketone testing, blood pressure, and body weight. More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 can be found in the IB.

4. Objectives and Endpoints

Table ITSI.2 shows the objectives and endpoints of the study.

Table ITSI.2. Objectives and Endpoints

Objectives	Endpoints
Primary Objective	
1. To compare LY900014 and insulin lispro with respect to the rate of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump	1. Rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump, during the 6-week treatment period
Secondary Objectives	
2. To compare LY900014 and insulin lispro with respect to the incidence of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump	2. Incidence (percent of patients with at least 1 event) of infusion set failures that lead to premature infusion set changes, due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump, during the 6-week treatment period
3. To compare LY900014 and insulin lispro with respect to the rate and incidence of premature infusion set changes	3. Rate (events/patient/30 days) and incidence of premature infusion set changes by reason (infusion set kinked, came out, or leaking; empty pump reservoir; infusion site pain or redness; pump occlusion alarm; suspected infusion set occlusion; other) during the 6-week treatment period
4. To compare LY900014 and insulin lispro with respect to the time interval until infusion set change	4. Time interval until infusion set change during the 6-week treatment period
5. To compare LY900014 and insulin lispro with respect to total, basal, and bolus insulin dose	5. Bolus/total insulin dose ratio at the end of the 6-week treatment period
6. To compare LY900014 and insulin lispro with respect to the interstitial glucose reduction rate from hyperglycemia following a non-meal-related correction bolus delivered via the pump	6. Interstitial glucose reduction rate (glucose reduction [mg/dL and mmol/L] per minute) within 4 hours following a non-meal-related correction bolus via the pump, from hyperglycemia (interstitial glucose >180 mg/dL (10.0 mmol/L) to recovery (interstitial glucose ≤180 mg/dL), from up to 6 weeks of CGM use
7. To compare LY900014 and insulin lispro with respect to the rate of severe hypoglycemic events	7. Rate (events/patient/100 years) of severe hypoglycemic events during the 6-week treatment period

Objectives	Endpoints
Tertiary/Exploratory Objectives	
8. To compare the safety of LY900014 and insulin lispro	8. Adverse events and vital signs
9. To compare LY900014 and insulin lispro with respect to the rate and incidence of documented symptomatic post-meal hypoglycemia	9. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with at least 1 event) of documented symptomatic post-meal hypoglycemia within 1 and 2 hours after the start of a meal during the last 2 weeks of the 6-week treatment period
10. To compare LY900014 and insulin lispro with respect to the rate and incidence of documented symptomatic hypoglycemia	10. Rate (events/patient/year and/or events/patient/30 days) and incidence (percent of patients with events) of documented symptomatic hypoglycemic events during the last 2 weeks of the 6-week treatment period
11. To compare LY900014 and insulin lispro with respect to the incremental AUCs after breakfast, obtained from CGM use	11. Incremental $AUC_{0\text{-}1\ hour}$ after breakfast during the last 2 weeks of up to 6 weeks of CGM use
12. To compare LY900014 and insulin lispro with respect to the duration of time glucose values are within target range (71 and 180 mg/dL [3.9 and 10.0 mmol/L]), obtained from CGM use	12. Duration (in minutes) and percentage of time with glucose values between 71 and 180 mg/dL (3.9 and 10.0 mmol/L), both inclusive, normalized to a 24-hour period, during the last 2 weeks of up to 6 weeks of CGM use
13. To compare LY900014 and insulin lispro with respect to the duration of time glucose values are within target range (71 and 140 mg/dL [3.9 and 7.8 mmol/L]), obtained from CGM use	13. Duration (in minutes) and percentage of time with glucose values between 71 and 140 mg/dL (3.9 and 7.8 mmol/L), both inclusive, normalized to a 24-hour period, during the last 2 weeks of up to 6 weeks of CGM use
14. To compare LY900014 and insulin lispro with respect to the glucose profiles, obtained from CGM use	14. Average glucose for a 24-hour period during the last 2 weeks of up to 6 weeks of CGM use
15. To compare LY900014 and insulin lispro with respect to the glucose variability, obtained from CGM use	15. Interquartile range, CV, LBGI, and HBGI during the last 2 weeks of up to 6 weeks of CGM use
16. To compare LY900014 and insulin lispro with respect to the factors affecting dosing in pumps	16. Actual and change from baseline in factors affecting dosing in pump (breakfast CR, AIT, breakfast ISF, and frequency of use of non-normal bolus type [Square Wave or Dual Wave]), during the 6-week treatment period
17. To compare LY900014 and insulin lispro with respect to the duration of time spent in hypoglycemic glucose ranges, obtained from CGM use	17. Duration (in minutes) and percentage of time with glucose values <50, <60, and ≤70 mg/dL (0.8, 3.3, and 3.9 mmol/L), normalized to a 24-hour period and number of episodes, defined as at least 10 consecutive minutes <50, <60, and ≤70 mg/dL, during the last 2 weeks of up to 6 weeks of CGM use

Objectives	Endpoints
18. To compare LY900014 and insulin lispro with respect to the duration of time spent in hyperglycemic glucose ranges, obtained from CGM use	18. Duration (in minutes) and percentage of time with glucose values >180, >250, and >300 mg/dL (10.0, 13.9, and 16.7 mmol/L), normalized to a 24-hour period and number of episodes, defined as at least 10 consecutive minutes >180, >250, and >300 mg/dL, during the last 2 weeks of up to 6 weeks of CGM use
19. To separately evaluate the glycemic control of LY900014 and insulin lispro	19. Summary statistics of actual and change from baseline to Week 6 in HbA1c for each treatment
20. To compare LY900014 and insulin lispro with respect to 1,5-AG	20. Actual and change from baseline to Week 6 in 1,5-AG

Abbreviations: 1,5-AG = 1,5-anhydroglucitol; AUC = area under the curve; CGM = continuous glucose monitoring; CV = coefficient of variation; HbA1c = hemoglobin A1c; HBGI = high blood glucose index; LBGI = low blood glucose index; SMBG = self-monitored blood glucose.

5. Study Design

5.1. Overall Design

Study I8B-MC-ITSI is a Phase 3, prospective, randomized, double-blind, outpatient, multinational, multicenter, 2-treatment group, crossover, active-controlled study conducted in patients with T1DM currently using CSII. In the 2 treatment groups, LY900014 and insulin lispro, meal bolus doses will be delivered immediately prior to each meal (0 to 2 minutes) in a double-blind manner.

The study is designed to compare LY900014 and insulin lispro with respect to the rate (events/patient/30 days) of infusion set failures that lead to premature infusion set changes due to a pump occlusion alarm OR due to unexplained hyperglycemia with blood glucose (SMBG) >250 mg/dL (13.9 mmol/L) that does not decrease within 1 hour following a correction bolus delivered via the pump. The study includes a 1-week screening period and a 2-week lead-in period followed by a 2-period crossover and a 4-week post-treatment safety follow-up. Each period of the crossover will consist of 6 weeks of treatment with no washout between periods.

Unexplained hyperglycemia is expected to be an uncommon event in this study and the use of real-time CGM, self-monitored blood glucose (SMBG), electronic clinical outcomes assessment (eCOA), frequent study visits, and provision of standard of care hyperglycemia management guidelines and blood ketone testing meters and strips will all serve to maximize subject safety. Patients will be experienced pump users who have recent experience using CGM or flash glucose monitoring (FGM), and they will be guided to follow best standards of care in insulin pump management.

Patients currently treated with a rapid-acting insulin analog via CSII will be eligible for inclusion in the trial. All patients treated with insulin aspart or insulin glulisine at screening will be transferred to insulin lispro at Visit 2 so that all patients are using insulin lispro during the lead-in period. Patients must be using the MiniMed 530G, MiniMed 630G (US) or 640G (EU) insulin pump. The bolus delivery speed for all pumps will be set to standard speed (1.5 U/minute) for the duration of the lead-in and treatment phases of the study. The purpose of the lead-in period (prior to randomization) will be to obtain preliminary diagnostic tests, determine baseline hypoglycemia rate, and evaluate pump basal rates and bolus calculator settings (carb ratio [CR], insulin sensitivity factor [ISF], and active insulin time [AIT] for calculating mealtime and correction bolus doses) for appropriateness. During the study, patients will be expected to use the pump's bolus calculator to determine mealtime and correction bolus doses. Patients will be required to suspend use of personal CGM or FGM and the SmartGuard/Threshold Suspend feature of their pump, if applicable, during the lead-in and treatment phases of the study.

Dexcom G5 will be used by all patients in real-time mode beginning at Visit 2 and continuing throughout the treatment phase of the study. Dexcom G5 will allow patients to see real-time continuous glucose readings every 5 minutes and can aid in the identification of trends and patterns in glucose levels and detection of hyperglycemia and hypoglycemia. Patient interpretation of the Dexcom G5 data should be based on the glucose trends and several

sequential readings over time. All patients will be required to have the alerts and alarm features of Dexcom G5 enabled throughout the study.

Patients will be required to perform SMBG using the study-provided glucometer a minimum of 4 times/day, including pre-meal and bedtime, for calibration of the Dexcom G5 Mobile Continuous Glucose Monitoring System (Dexcom G5) every 12 hours, and to confirm episodes of hypoglycemia, unexplained hyperglycemia, and the SMBG response to a correction bolus.

At every office visit, the date and time of each device used in the study (insulin pump, eCOA diary, glucometer, ketone meter, and CGM receiver) should be synchronized and verified by site staff.

At Visit 3 (Randomization Visit), patients will be randomly assigned to 1 of the 2 treatment sequences of double-blind LY900014 and insulin lispro. During Visit 3, under the observation of site staff, patients will insert a new CGM sensor and begin a new sensor session. Also under the observation of site staff, patients will fill a new pump reservoir and infusion set with investigational product (IP), insert the infusion set cannula into an appropriate pump site, and begin infusion of IP.

At Week 6 (Visit 6), patients will cross-over to the alternate treatment. Investigators will reset the patient's pump to the bolus calculator settings (CR, ISF, and AIT) and basal rates that had been optimized during the lead-in period. During the visit, under the observation of site staff, patients will insert a new CGM sensor and begin a new sensor session. Also under the observation of site staff, patients will fill a new pump reservoir and infusion set with IP, insert the infusion set cannula into an appropriate pump site, and begin infusion of IP.

During the treatment periods, patients and investigators will interact frequently and use SMBG and CGM data to optimize pump settings in order to maximize glycemic control.

The overall glycemic control goals for all patients enrolled in the study are similar to those recommended by the American Association of Clinical Endocrinologists (Bailey et al. 2016). The glycemic targets for this study are for patients to achieve a pre-prandial glucose level of approximately 100 mg/dL (5.6 mmol/L), a 1- to 2-hour PPG level of <180 mg/dL (<10.0 mmol/L), and a bedtime glucose level of approximately 90-130 mg/dL (5.0-7.2 mmol/L) with no hypoglycemia (defined as BG ≤70 mg/dL [3.9 mmol/L]).

During the first 4 weeks of each treatment period, patients, in consultation with investigators, should use SMBG, CGM, and hypoglycemia data to adjust basal rates and bolus calculator settings (CR, ISF, and AIT) as necessary to meet the glycemic targets.

During the last 2 weeks of each treatment period, it is expected that CR, ISF, AIT, and basal rates would remain fairly stable, with adjustments if needed for safety reasons, such as hypoglycemia or unacceptable hyperglycemia. It is important to keep these as stable as possible, because CGM and hypoglycemia data from this period will be used for treatment comparisons.

Study governance considerations are described in detail in Appendix 3.

Figure ITSI.1 illustrates the study design.

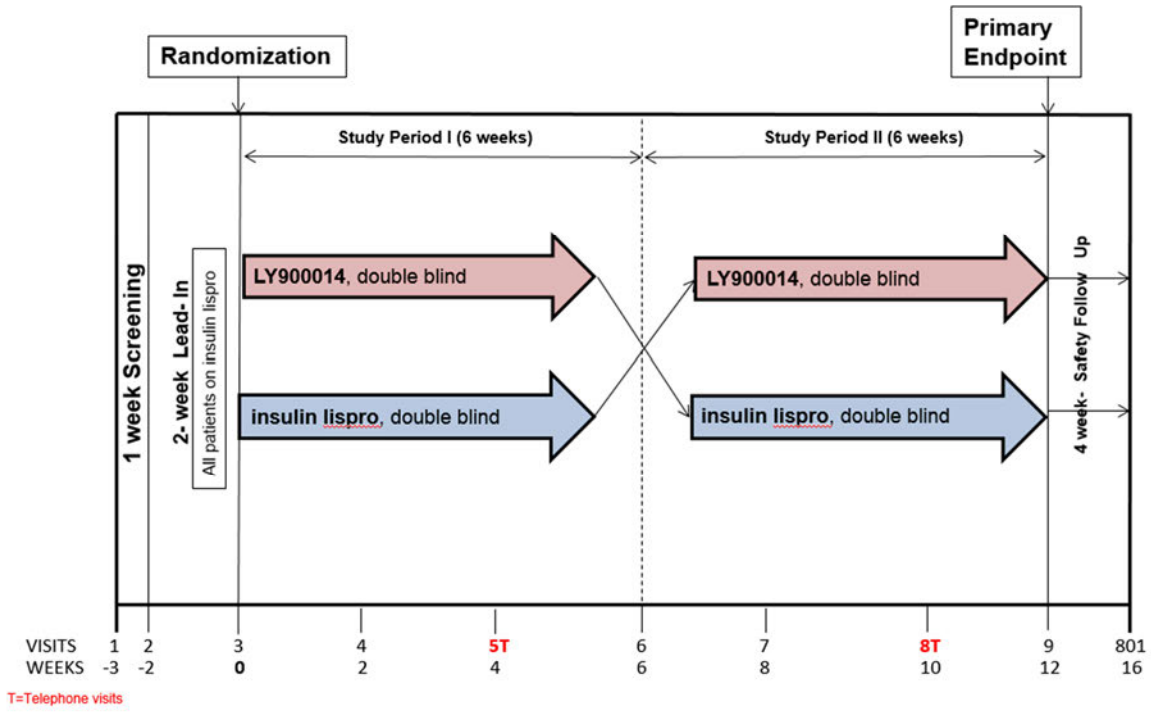


Figure ITSI.1. Illustration of study design for Clinical Protocol I8B-MC-ITSI.

5.2. Number of Participants

Approximately 60 participants will be screened to achieve 48 randomized patients for a total of approximately 42 patients to complete the study.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

A crossover design was chosen for this study to reduce variability and increase the power as each patient serves as his or her own control.

At the crossover point, patients are required to change their reservoir and infusion set, insert a new CGM sensor, and begin a new sensor session, and investigator will reset the patient's pump to the settings that had been optimized during the lead-in period. This avoids carryover effects on catheter occlusion-related treatment comparisons.

In addition, patients are encouraged to reach stable dose during the last 2 weeks of each period. This also minimizes carryover effect on the exploratory safety and efficacy outcomes.

5.5. Justification for Dose

Insulin lispro has been in the market for over 20 years with widespread use worldwide and has extensive efficacy and safety data. LY900014 will have the same insulin lispro concentration

(100 U/mL) as that of commercially available Humalog. The addition of treprostinil to the insulin lispro formulation does not modify the physical, chemical, or biological integrity of insulin lispro. The total serum insulin lispro exposure (area under the concentration curve from 0 to 5 hour postdose) are similar between LY900014 and insulin lispro in T1DM CSII. In addition, the total glucodynamic activity as measured in euglycemic clamp studies was similar between LY900014 and Humalog. In previous studies, these insulins were substituted unit for unit. The basal rates and bolus doses of insulins used in this study should be determined based on the individual needs of each patient.

6. Study Population

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted. Individuals who do not meet the criteria for participation in this study (screen failure) may not be rescreened. Retests are also not allowed, except for cases when results are not available from the original sample.

6.1. Inclusion Criteria

Patients are eligible to be included in the study only if they meet all of the following criteria at screening:

Type of Patient and Disease Characteristics

[1] Men or women diagnosed (clinically) with T1D, based on the World Health Organization classification (Appendix 5) for at least 1 year prior to screening, and continuously using insulin for at least 1 year

Patient Characteristics

[2] Are at least 18 years of age

[3] Male patients:

a) No male contraception required except in compliance with specific local government study requirements.

[4] Female patients:

a) Women not of childbearing potential may participate and include those who are:

i) infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as Mullerian agenesis

Or

ii) postmenopausal – defined as either

(1) a woman 50 to 54 years of age (inclusive) with an intact uterus, not on hormone therapy who has had either

(a) cessation of menses for at least 1 year

Or

(b) at least 6 months of spontaneous amenorrhea with a follicle-stimulating hormone >40 mIU/mL

Or

(2) a woman aged 55 years or older not on hormone therapy, who has had at least 6 months of spontaneous amenorrhea

Or

(3) a woman of at least 55 years of age with a diagnosis of menopause prior to starting hormone replacement therapy

b) Women of childbearing potential participating:

i) cannot be pregnant or intend to become pregnant

ii) cannot be breastfeeding (including the use of a breast pump)

iii) must remain abstinent or use 1 highly effective method of contraception or a combination of 2 effective methods of contraception for the entirety of the study (Appendix 6)

iv) test negative for pregnancy at the time of screening (Visit 1); note: a urine pregnancy test is conducted at Visit 3

[5] Have been using CSII therapy for a minimum of 6 months prior to screening. Interruption of CSII is allowed once during the 6 months prior to screening for up to 14 consecutive days, such as during a hospitalization, a pump malfunction, or a "pump holiday"

[6] Are currently treated with the same rapid-acting analog insulin (insulin lispro U-100, insulin aspart, or insulin glulisine) via CSII (where approved) for at least the last 30 days prior to screening

[7] Have been using CGM or FGM (CGM preferred) for a total of at least 60 days during the 12 months prior to screening

[8] Have HbA1c values ≤8.5%, as determined by the central laboratory at screening (Visit 1)

[9] Have a body mass index (BMI) of ≤35 kg/m^2 at screening (Visit 1)

[10] Are proficient, in the opinion of the investigator, in carbohydrate counting, and in adjusting basal rates and bolus calculator settings (CR, ISF, and AIT)

[11] Are willing to take at least 3 meal-time insulin boluses every day on a regular basis throughout the trial

[12] Are willing to use the pump's bolus calculator (Bolus Wizard) to determine mealtime and correction bolus doses

[13] Are willing to adhere to an infusion set and reservoir change interval of every 72 ± 4 hours unless a change is required for failure of the infusion set

[14] Are willing to perform SMBG, using the study glucometer, at least 4 times daily (including every 12 hours to calibrate the sensor), and to confirm hypoglycemia, unexplained hyperglycemia, and BG response following a correction bolus delivered via the pump

[15] Are willing to use the Dexcom G5 Mobile Continuous Glucose Monitor in real-time mode throughout the study

[16] Are willing to suspend use of personal CGM or FGM throughout the study

[17] Are willing to suspend use of SmartGuard/Threshold Suspend throughout the study

[18] Are willing to avoid use of all products containing acetaminophen or paracetamol while using the Dexcom CGM system

[19] Must be using a MiniMed 530G or 630G (US) or 640G (EU) insulin pump for at least the last 30 days and willing to stay on the same pump throughout the study (pump must be approved for use in the country where the patient resides)

[20] Must use study-provided MiniMed insulin pump reservoirs and Mio infusion sets

[21] Are willing to keep records in eCOA devices as required by this protocol

[22] Have access to a telephone, or alternative means for close monitoring and communications, and have access to a reliable cellular signal or wireless internet access for transmission of the eCOA data

[23] Have refrigeration in the home or have ready access to refrigeration for storage of insulin

Informed Consent

[24] Have given written informed consent to participate in this study in accordance with local regulations

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening:

Medical Conditions

[25] Have hypoglycemia unawareness as judged by the investigator

[26] Have had more than 1 episode of severe hypoglycemia (defined as requiring assistance due to neurologically disabling hypoglycemia) within 6 months prior to screening

[27] Have had more than 1 emergency room visit or hospitalization due to poor glucose control (hyperglycemia or DKA) within 6 months prior to screening

[28] Have significant insulin resistance defined as having received a total daily dose of insulin >1.2 U/kg at the time of screening (Visit 1), as determined by the average total daily insulin dose over the 3 days prior to screening divided by weight in kilograms based on investigator review of the patient's pump history

[29] Have significant lipohypertrophy, lipoatrophy, or scars within the subcutaneous tissue in areas of infusion, in the opinion of the investigator

[30] Have a history of abscess at an infusion site within the last 90 days prior to screening

[31] Have a history within the past 6 months of changing their pump infusion set more frequently than every 3 days on a regular basis

[32] Have vision loss or hearing loss that does not allow recognition of pump screens, alerts, and alarms

[33] Have presence of clinically significant gastrointestinal disease (for example, clinically active gastroparesis associated with wide glucose fluctuations; in the investigator's opinion) or gastric bypass.

[34] Have any other condition (including known drug or alcohol abuse or psychiatric disorder including eating disorder) that precludes the patient from following and completing the protocol

[35] Have cardiovascular disease, within the last 6 months prior to screening, defined as stroke, decompensated heart failure New York Heart Association class III or IV (Appendix 7), myocardial infarction, unstable angina pectoris, or coronary arterial bypass graft

[36] Renal:

a. History of renal transplantation

b. Currently receiving renal dialysis

c. Serum creatinine >2.0 mg/dL (177 µmol/L) at screening (Visit 1) as measured by the central laboratory

[37] Hepatic:

a. Have obvious clinical signs or symptoms of liver disease (for example, acute or chronic hepatitis, or cirrhosis), or elevated liver enzyme measurements as indicated below at screening (Visit 1)

b. Total bilirubin level (TBL) ≥2X the upper limit of normal (ULN) (with the exception of Gilbert's disease) as defined by the central laboratory Or

c. Alanine aminotransferase (ALT) or aspartate aminotransferase (AST) ≥3X ULN as defined by the central laboratory

[38] Malignancy: have active or untreated malignancy, have been in remission from clinically significant malignancy (other than basal cell or squamous cell skin cancer) for less than 5 years, or are at an increased risk for developing cancer or a recurrence of cancer in the opinion of the investigator

[39] Have any hypersensitivity or allergy to any of the insulins or excipients used in this trial

[40] Hematologic: have had a blood transfusion or severe blood loss within 90 days prior to screening or have known hemoglobinopathy, anemias, or any other traits known to interfere with measurement of HbA1c

Prior/Concomitant Therapy

[41] Have used insulin human inhalation powder (Afrezza®) within 90 days prior to screening

[42] Are receiving any oral or injectable medication intended for the treatment of diabetes mellitus other than rapid-acting analog insulin via CSII in the 90 days prior to screening. Occasional pen or syringe injection of insulin is allowed in the event of pump malfunction, unexplained hyperglycemia not responsive to pump correction bolus, etc.

[43] Glucocorticoid therapy: are receiving chronic (lasting longer than 7 consecutive days) systemic glucocorticoid therapy (including intravenous, IM, subcutaneous, and oral) or intra-articular, but excluding topical, intraocular, intranasal, and inhaled preparations), or have received such therapy within 8 weeks immediately prior to screening with the exception of replacement therapy for adrenal insufficiency

Prior/Concurrent Clinical Trial Experience

[44] Are currently enrolled in any other clinical study involving an IP or any other type of medical research judged not to be scientifically or medically compatible with this study

[45] Have participated, within the last 30 days in a clinical trial involving an IP. If the previous IP has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed

[46] Have previously completed or withdrawn from this study after having signed the informed consent form (ICF) or any other study investigating LY900014 after receiving at least 1 dose of the IP

Other Exclusions

[47] Have an irregular sleep/wake cycle (for example, patients who sleep during the day and work during the night)

[48] Are investigator site personnel directly affiliated with this study and/or their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted

[49] Are Lilly employees or representative (including employees, temporary contract workers, or designees responsible for the conduct of the study)

[50] Are unable and/or unwilling to provide informed consent, to make themselves available for the duration of the study, or to abide by study procedures

6.3. Lifestyle Restrictions

Study participants should be instructed not to donate blood or blood products during the study.
Patients should be instructed to avoid major changes in diet or exercise during the study.
Patients should be instructed to avoid all medications that contain acetaminophen/paracetamol during the study.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be rescreened. Retests are also not allowed, except for cases when results are not available from the original sample.